[External] Scientific Reports: Decision on "Exosite inhibition of ADAMTS-5 by a glycoconjugated arylsulfonamide"

Scientific Reports <do-not-reply@springernature.com>
Wed 8/5/2020 9:35 AM
To:Mehta, Akul Y (BIDMC - Cummings - General Surg SF) <aymehta@bidmc.harvard.edu>

Dear Dr Mehta,

Thank you for your help with the manuscript, "Exosite inhibition of ADAMTS-5 by a glycoconjugated arylsulfonamide", which you recently reviewed for Scientific Reports.

For your records, the decision on this manuscript, based partly on your input, was: Revise. Any comments to authors have been appended below.

We greatly appreciate your assistance and participation in the review process for Scientific Reports and hope that we can continue to benefit from your expertise on future submissions.

Kind regards,
Peer Review Advisors
Scientific Reports

Reviewer 1
The authors present a discovery of "exosite inhibition of ADAMTS-5 by glycoconjugated arylsulfonamide", which could potentially open up a new class of inhibitors for osteoarthiritis. The paper is well written and majority of the data is presented in a robust manner. The work is of interest and should be recommended for publication provided the authors address the comments below.

1. Is there a known propensity for these molecules to form micelles or undergo aggregation at concentrations where inhibition is observed? If tested please report your findings, and if not then please check if it can (at least for lead compound 4b). Given the ampiphilic nature of the molecule (polar head - GlcNAc and non-polar tail sulfonamide) this could be a possibility and could skew results. Please report the critical micelle concentration, if these compounds do form micelles. I advise the authors to read this chapter on assay interference by aggregation https://urldefense.proofpoint.com/v2/url?u=https-3A__www.ncbi.nlm.nih.gov_books_NBK442297_&d=DwIFaQ&c=WknmpdNpvrlj2B5K1aWVqL1SOiF30547pqSuOmtwXTQ&r=1NcKsZ95GZXswgWo6pK57YveyNDcCDLZdRnxJDzG6_c&m=IXdEfOw2O6KrxXeRXRnj2S35uP8EQT_PAJoP-bmybn0&s=WWdGAxVmwfr67CBhakWoZboTbZeRis0UYfZKn3LUAn8&e=
2. The authors must provide the IC50 curves with all data points, from which the IC50 values in Table 2 are calculated in the supplementary. This is important as the curve structure can potentially be useful to judge inhibitory parameters for allosteric/exosite inhibitors e.g. Hill slope.
3. Major Concern: Replacement of the sugar moiety with benzoyl group (compound 5) is far from ideal for comparison of the importance of the GlcNAc. The authors should try to replace GlcNAc with GalNAc (single hydroxy configuration change) and/or ManNAc, which would be more relevant comparison. This needs to be done and used as a control in at least the inhibition studies shown in Table 2 and the western blot studies.
4. Table 1 & 2 shows NI as "not inhibiting at 50 µM". What percentage limit was used as a the cutoff, i.e. 10% of control or 20% of control or 50% of control? Please clarify if it means "not inhibiting >10% at 50 µM".

5. Major Concern: Please perform the molecular docking on at least two crystal structures if available. The authors have performed the study on PDB id: 2RJQ but provided no justification as to why this crystal structure was chosen. This would act as an internal control for the robustness of the method for this system.
6. Does the docking also show similar results if GM6001 is removed from the site, i.e. uninhibited protein structure if available?
7. Figure S2 legend needs to define what is the structure shown in purple stick model i.e GM6001. Figure S2 legend also needs to define the zinc grey sphere.
8. The authors should make the docked poses available as supplementary files for independent analysis.
9. The authors should also try docking/substituting other structures in the series, for e.g. compound 4c, to compare the binding pose and justify why a difference of ~6 fold is observed with substituents on the R1 benzene ring.
10. Major Concern: Figure 3C data points for the compound 4b curve misses important points between 100-400 nM of GM6001 which are important from a curve fitting standpoint. Please redo this experiment and the control and show complete data for all data points. Similarly panel 3D does not show data point for compound 4b curve at 0 nM of GM6001, and it is not clear if this point exists for panel 3C either. The 0 nM GM6001 data point needs to be calculated for compound 4b alone in comparison to the DMSO control at 0 nM GM6001 data point. This is important to show that in this assay, independent of GM6001, compound 4b was still able to produce close to 50% inhibition (or based on your curves close to 35%) as indicated by the IC50. Out of everything these experiments absolutely feel like they need to be redone to get robust results with all data points. In addition please redo this experiment at concentration of 4b at 5 μM (slightly lower than IC50) and 15 μM (slightly higher than IC50). This is important to judge true synergism between 4b and GM6001.
11. Figure 3 legend also refers to "panel G" which is a typo as there is no panel G.
11. Please provide all details of protein expression in this manuscript, even if it was done as previously described.

Overall the work is good and the only major issue I see is with point number 3, 5 and 10 which I feel the authors must address prior to qualification for publication.

Reviewer 2
The manuscript by Salvatore et al describes the identification of new ADAMTS-5-targeting agents. The protein appears to be a viable target to develop disease-modifying agents that may help in the management of diseases like osteoarthritis. In this direction, the authors intuitively designed glycoconjugated arylsulfonamide derivatives. The design was partially guided by glycosaminoglycans which also target ADAMTS-5. The authors put forward the molecule 4b to introduce the concept of ADAMTS-5 exosite-mediated inhibition. The molecule has also been presented as the best candidate for future development. The molecule activity, however, is mediocre (~10 μM). Nevertheless, the concept is novel as the authors have attempted to engineer functional selectivity toward ADAMTS-5 by targeting a previously unstudied exosite.

The authors used a series of chemical, biochemical, and computational experiments to verify the in vitro activity of 4b and its binding site. It appears that all methods relevant to the chemical synthesis and characterization (NMR, melting points, and elemental analysis), biological testing (kinetics and western blotting), and molecular modeling and dynamics have properly been conducted. All experimental details are provided. The putative binding site has been largely verified by computational tools as well as mutagenesis studies. The statistical analysis is sufficient and there are no ethical concerns regarding the use of animals or human subjects.

Overall, the manuscript is concise and adequately organized. I recommend accepting the manuscript after addressing the following: a) the statement in page 4 “The compound was docked into the catalytic site of ADAMTS-5” is confusing given the predicted binding site of the molecule and b) please re-write/clarify the legend for figure 3.

Our flexible approach during the COVID-19 pandemic

If you need more time at any stage of the peer-review process, please do let us know. While our systems will continue to remind you of the original timelines, we aim to be as flexible as possible during the current pandemic.